

AMIT JHA

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EDUCATION

Indian Institute of Technology (IIT), Jodhpur

Jodhpur, India

Bachelor of Technology in Computer Science & Engineering; GPA: 9.1/10.0

July 2019 – May 2023

- **Coursework:** Linear Algebra, Probability & Statistics, Numerical Methods, Data Structures & Algorithms.
- **Awards:** Institute Merit-cum-Means Scholarship recipient (Top 5% of CSE department).

World Quantitative Academy (WQA)

Remote

Advanced Certificate in Quantitative Finance & Volatility Modeling

June 2023 – Nov 2023

- Intensive coursework in Stochastic Calculus, Black-Scholes PDE discretization, and Monte Carlo methods.

PROFESSIONAL EXPERIENCE

Model Risk & Quant Solutions Group

Mumbai, India

Quantitative Analyst / Model Validator

June 2023 – Present

- Perform rigorous mathematical validation on sell-side front-office pricing engines (Equity, FX, and Interest Rate).
- Validated Crank-Nicolson finite difference schemes for barrier options, verifying stability bounds under Rannacher time-stepping.
- Audited stochastic volatility models (Heston, SABR) by verifying asymptotic wing behaviors and local vol Dupire mappings.
- Developed independent Python libraries to replicate Monte Carlo paths using the Quadratic Exponential (QE) scheme.

High Frequency Trading Lab, IIT

Jodhpur, India

Quantitative Research Intern

May 2022 – July 2022

- Designed and backtested intraday statistical arbitrage strategies on NSE equities using order book imbalance features.
- Optimized a C++17 trading simulation engine using vectorization (Eigen library), reducing simulation latency by 35%.
- Evaluated cointegration relationships (Johansen test) and calibrated mean-reversion rates using Ornstein-Uhlenbeck processes.

SELECTED PROJECTS

High-Performance Equity Options Pricing Engine (C++17)

[GitHub Link](#)

- Developed a modular PDE solver using Finite Difference Methods (Explicit, Implicit, and Crank-Nicolson schemes).
- Implemented SOR (Successive Over-Relaxation) and Thomas algorithms in C++ for tridiagonal matrix inversion.
- Wrote custom Python bindings using pybind11 to visualize options Greeks surface grids in Jupyter notebooks.

Heston Model Volatility Calibration (Python, SciPy, PyTorch)

[GitHub Link](#)

- Calibrated Heston model parameters ($v_0, \kappa, \theta, \sigma, \rho$) to market-implied volatility surfaces.
- Implemented the Carr-Madan FFT (Fast Fourier Transform) option pricing methodology using characteristic functions.
- Structured a deep PINN (Physics-Informed Neural Network) solver to compute option values directly from the BS PDE.

TECHNICAL SKILLS

- **Languages:** C++ (17/20, STL, Template Programming), Python (NumPy, SciPy, Pandas, PyTorch), SQL, Bash, LaTeX.
- **Quantitative Finance:** Black-Scholes PDE, Stochastic Volatility (SABR, Heston), Finite Differences, Monte Carlo.
- **Libraries & Tools:** Eigen, Boost.Asio, Git, Docker, Linux (Ubuntu/CentOS), Visual Studio Code, CMake.